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An Integrated Approach for Scheduling and Validation of Steady-State Production Well Tests in Unconventional Wells

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Abstract

Steady-state production well tests are a key component of day-to-day operations, providing critical data for regulatory reporting, volume allocation, and identification of well-work opportunities. This paper presents an automated workflow deployed for scheduling and validating steady-state production well tests across more than 2,300 ExxonMobil-operated wells in the Permian Basin.

A Well-Test Scheduling (WTS) workflow was developed to provide daily recommendations to field personnel for prioritizing well tests. It leverages data such as well-to-separator mappings, recent well-test history, and ongoing well events such as frac hits or shut-in events. In facilities equipped with automated routing headers, recommendations are implemented via SCADA remote setpoint control, with adjustable test and purge durations.

Upon test completion, an automated Well-Test Validation (WTV) program ensures test quality using high-frequency data and a rule-based validation system. This system assesses the quality of separation, infers the effective flow period, and verifies the measured oil, water, and gas rates. Results are communicated to field operators through a mobile app that is integrated with production accounting systems.

The WTS process has improved well-test prioritization, ensuring at least one well test per month and increased testing frequency on a per-well basis. The well-test validation process has enhanced the accuracy of oil, water, and gas rate measurements, aiding daily surveillance by promptly detecting issues like water carryover, sand buildup in test separators, and water-in-oil emulsions. The automated process has reduced test turnover time from 4 days to less than 2 hours, significantly saving time for field operators and engineering teams while increasing test acceptance from 65% to 75%. Finally, the WTV process has established a comprehensive dataset linking flowing pressures with phase rates, laying the groundwork for machine learning applications in surveillance and optimization.

Keywords: Well-Test Validation, Production Well Tests, Permian Basin, Scheduling Well Tests

Introduction

Steady-state production well testing is an integral operational activity in any oil and gas field and is nearly essential for meeting regulatory compliance, identifying production optimization opportunities, and improving our understanding of reservoir and completion performance. In unconventional wells, frequent and reliable production well tests provide the information necessary to delineate the steep declines and historize the production rates necessary for optimization and future reservoir development.

The scale of modern unconventional fields combined with, what is often, unfavorable well-to-personnel ratios and the ever-increasing need for reliable and representative well rate measurements impose the need to streamline the scheduling and validation process of production well tests. In this work, we present the Well Test Scheduling and Validation workflows that have been implemented successfully across more than 2,300 operated wells in the Permian Basin.

Conventional Practice and Challenges

Historically, well testing has been, for the most part, a manual process. The flow-stream of a well is routed to a dedicated test separator where oil, water, and gas volumes are measured for a predetermined duration, typically for 24 hours. These volumes are recorded by operators or field equipment, prorated to daily rates for durations shorter than 24 hours, and stored for later use of allocating field production and facility incurred costs to individual wells. To optimize field production (e.g. Gas Lift Optimization) and quickly respond to curtailment events (e.g. third-party gas takeaway upsets), phase ratios need to be accurate and up to date, imposing the need for frequent well tests, well beyond the requirements imposed by federal and/or state regulations, particularly for new wells. In other words, frequent and reliable well tests are key to optimal operation of complex unconventional fields under constraints.

Under conventional well-testing practices, field operators typically require one to four days to manually approve a production well test after its completion. This delay can lead to short-term suboptimal production outcomes as closed-loop optimization algorithms rely exclusively on officially approved well-test data. Although manual approval satisfies regulatory requirements—such as monthly or quarterly reporting—a small subset of well-test records may remain unprocessed in a "Pending Review" status. Such delays hinder operational responsiveness and highlight the need to reduce processing times and ensure that all tests are evaluated. To address this, consistent stewardship is essential to guarantee that production well tests are promptly reviewed and accurately approved or rejected for allocation purposes.

Quality of Rate Measurements and Utilization of Field Equipment

Prior practices in executing well tests have created occasional erroneous entries of prorated volumes. For example, in certain, yet limited, instances the volume of an 8-hour test was incorrectly assigned as the daily rate, significantly underestimating the well's flow potential and impacting near-term optimization decisions. The data quality gets more complicated in gas-lifted wells where field operators need to validate both the total produced gas and the injected gas rates to accurately estimate net (formation) gas production and gas-to-oil ratios.

Well testing protocols were previously focused on meeting regulatory reporting requirements rather than collecting daily information to realize value from production optimization opportunities. For example, a test separator routed to a single well only recorded three well tests over the course of a month which is sufficient to meet the regulatory requirement but insufficient to meet daily optimization objectives. On a separate occasion, a shut-in well was routed to a test separator over the course of multiple days, preventing flowing wells from being tested and resulting in suboptimal data gathering practices and equipment utilization.

The scale of modern unconventional fields combined with unfavorable well-to-personnel ratios and the ever-increasing need for reliable and comprehensive well rate measurements impose the need to streamline the scheduling and validation process of production well tests. The extensive checks to validate tests (Engelman, 1968) add to the significant workload of field operators and, in conjunction with the potential

for incorrect assignments, suggest the need for a systematic and automated validation process. In this work, we present the Well-Test Scheduling and Validation workflows that have been implemented successfully across more than 2,300 operated wells in the Permian Basin, USA.

Field Instrumentation and Well-Testing Procedure

Permian Basin wells typically produce fluids containing vapor and liquid hydrocarbons phases and water. Produced fluids are routed, via a well header (Figure 1), to either a three-phase production (bulk) separator or a three-phase test separator (Figure 2) where a single well is tested. Patra et al. (2022) have provided additional context on well-testing procedures for commingled well set-ups, but the current study targets individual well testing.



Figure 1—Well header for routing wells to test separators or bulk separators



Figure 2—Test separators for unconventional wells

A standard 3-phase separator is configured such that the produced fluid is separated into gas, water, and liquid hydrocarbons using gravity separation. The gas phase is extracted from the top of the separator and measured using an orifice meter. The hydrocarbon liquid phase is separated from the water over a weir with the water being extracted from the bottom of the separator and measured with a magnetic or turbine meter. Under normal separation conditions, the oil phase flows over the weir into the ‘oil leg’ and is typically measured using a Coriolis meter. Coriolis meters, as compared to either orifice or magnetic meters, provide mass flow rates and additional measurements (AL-Mubarak, 1996) such as fluid density or drive gain (an indicator of gas in the oil leg) which is key in monitoring separation efficiency, as will be discussed later.

For gas-lift injected wells, an additional gas rate measurement for the lift gas is required to determine the reservoir gas/oil ratio and is measured using an orifice meter on the gas injection line near the wellhead.

A steady state production well test is initiated by physically routing a well into a test separator through the manual well-header valve (Figure 1) or through an automated actuating valve. Permian Basin operations have elected to install Human Machine Interfaces (HMI) at the separators to streamline the capture of well-test information, enabling the operator to enter the well on test directly using the HMI with pre-determined test durations, typically 8, 12 or 24 hours. At the end of the test, field automation equipment captures the accumulated volumes over the test period. If the well is not swapped out for a new well at the end of the specified test duration, a new test may automatically commence following the completion of the prior test. For each well test, oil, water, and gas volumes are reported by the field automation equipment and published to cloud databases as described in the following section. The field operators can prematurely terminate (i.e. stop and capture) or abort (i.e. discard) a well-test record, if necessary.

Upon the completion of a well test, the produced oil, water, gas, and gas injection volumes are made available to a mobile application where field operators can review and subsequently approve or reject a production well test by comparing the current well performance to prior well tests and considering any facility or well events that occurred during the well test.

Field Data Collection

The automated Well Test Validation (WTV) process is comprised of loosely coupled Operational Technology (OT) as well as Information Technology (IT) processes and systems. The data originates on the edge of the network typically in some form of a controller such as a Programmable Logic Controller (PLC) or Remote Terminal Unit (RTU) that communicates with field devices, sensors and electronic flow measurement (EFM) equipment. This data is collected into the controller and runs through either standard third-party well-test programs or internal well-test logic developed by field personnel.

A typical well-test data gathering program, as shown in Figure 3, has a set of registers to store ongoing and completed well tests as well as to notify other systems that a well test has completed. Edge compute resources also run at these locations which connect to and monitor the PLC controller over various OT protocols. Processes on these edge devices then capture the current well-test results and send them to middleware utilizing the Message Queuing Telemetry Transport (MQTT) protocol. Subscribing nodes within the SCADA system then monitor the MQTT data feeding into the MQTT brokers, capture any well-test records, and store this information to a centralized SQL database. The local Edge compute devices continually publish high-resolution, real-time EFM volumes and associated test vessel data during each well test over MQTT. The subscribing SCADA system historizes this data in both on-prem databases and the cloud where it is housed as time-series data and utilized in WTV analytics. A secure HTTPS rest API endpoint is also available from on premises as well as the cloud to request well-test records over a specified time frame. This API can be used both to deliver well-test atomic record information (referred to as ‘raw’ well-test records that provide information on the well being tested, test duration, start time, volumes produced, etc.) as well as receive feedback from the WTV workflow with regards to the validity of any given well test for ingestion in downstream systems and ultimately for display to the end users in the field.

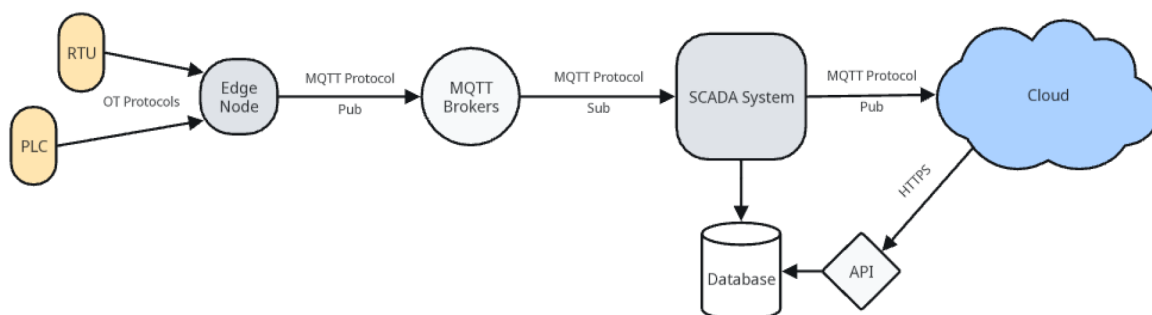


Figure 3—Field Data architecture

The data collection process for high-frequency field SCADA data, such as measurements from wellhead, downhole or surface facility instrumentation has been described in detail in prior publications (Movahed et al. (2024), Wang et al. (2024))

Data QAQC

To ensure that engineering workflows utilize high-quality field measurements, an automated Quality Assurance and Quality Control (QAQC) workflow was developed and implemented. The QAQC workflow has been discussed in detail elsewhere (Movahed et al. (2024), Kumar et al. (2025)) and focuses on the following tasks:

Source Selection: Selects measurement data that has high availability and reliability amongst all available sources for a given measurement/instrument. For example, at the Coriolis meter, oil volume accumulation measurements may be sporadic (continuous accumulation) but the volume accumulation with daily reset may have a higher resolution. In this case, volume accumulation time series is used for validating results.

Well-Instrument mapping integrity: Utilize cross-measurement correlations to ensure that high-frequency sensor data is reliable and appropriately mapped to the well of interest. For example, uncorrelated wellhead and downhole pressure measurements may indicate a potential swapped sensor tag between nearby wells.

Anomaly Detection: Select measurements that do not have outages, static points (i.e. flatlines) and/or unexpected trends.

Producing Method Validation: Validate that the well's current producing method in record systems is consistent with the observed high-frequency field measurements. Accurately capturing the well's producing method is key in performing the necessary checks for the approval or rejection of a well test.

Well Test Scheduling

Scheduling of production well tests is typically performed by the operations teams on a weekly basis, prioritizing regulatory compliance and requests from the engineering teams for upcoming well work or rate variance resolution from prior tests. Occasionally, the scheduling may be revised during daily operations meetings. For test separators equipped with automated routing headers, the wells are sequentially rotated for daily well tests. The weekly planning of well tests and the sequential rotations of wells in automated headers has been shown to be inefficient due to rapidly changing field conditions (e.g. downtime or production inhibiting events) imposing the need for a scheduling workflow that aims to improve equipment utilization, increase well testing frequencies and maximize data collection efficiencies for unconventional wells.

A proprietary non-linear optimization model was developed that furnishes daily recommendations on well-test scheduling for each test separator in the field. The model recommends, for each test separator, a hierarchy of wells to be tested ensuring regulatory compliance and accommodating, if possible, any desired well-testing frequencies provided by engineering teams (e.g. post-well-work productivity monitoring). This hierarchy includes up to three recommended wells per test separator in order of priority, providing field

operators with the ability to choose a second/third option, if the first recommended well cannot be tested due to any reason. The model considers various parameters such as well age, producing liquid rate, and recent well-test history and is integrated with near-real-time surveillance of production inhibiting events to ensure measurement of normal representative production rates. For example, wells incurring a frac-hit (which are events when a nearby well is undergoing a hydraulic fracture treatment that impacts – usually negatively – an older well that has already been producing) are prioritized for testing to accurately delineate the rapidly changing phase ratios, enabling both improved oil/water liquid allocations and optimized well controls during curtailment events (e.g. water takeaway). Conversely, wells experiencing downtime events are deprioritized, leaving room for actively producing wells to be tested. The recommendations are furnished to the operation teams via dashboards or SMS cellular service and, in facilities with automated headers the recommendations are implemented in a closed-loop fashion with flexibility to adjust both the test and purge durations, if necessary. In every case, field operators maintain full control of determining the well on test and can overwrite recommendations as dictated by field conditions. Figure 4 illustrates the logic of the well-test scheduling workflow.

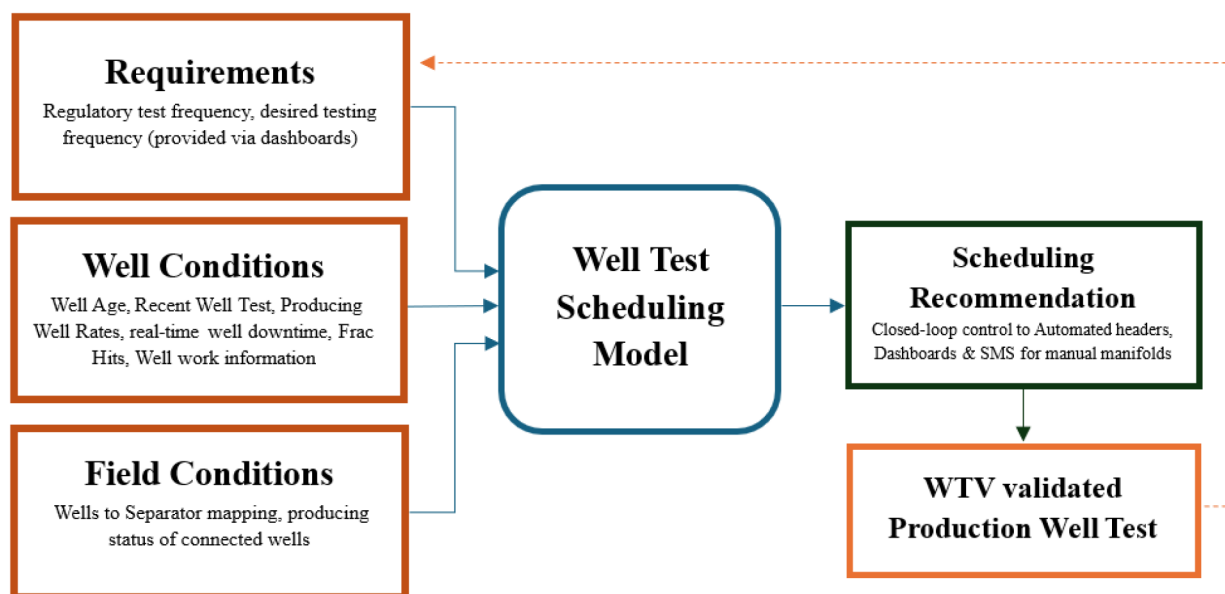


Figure 4—Input and output of the Well-Test Scheduling workflow

The well-test scheduling workflow has been successfully deployed at scale across operated fields in the Midland Basin. Figure 5 illustrates the desired versus actual well-test counts for a test separator with flowline connections to 9 wells over the course of a month. Such a high well-to-separator ratio is not typical but can be used to illustrate the results of the workflow. As is shown, 6 out of the 9 wells met and/or exceeded the desired well-testing frequency for the month. Figure 6 presents a case involving a test separator with an automated header. The workflow initially prioritized well tests for Well 1, targeting 20 tests for the month. However, due to a 5-day shutdown of Well 1, the system dynamically reallocated testing to the remaining wells to maintain operational efficiency. Overall, the scheduling recommendations have been well received by field operations personnel, resulting in a user adoption rate of 86% of the daily recommendations (see Figure 7). The results demonstrate a clear positive impact of the technology in optimizing the utilization of existing surface equipment and improving operational efficiency from scheduling to execution.

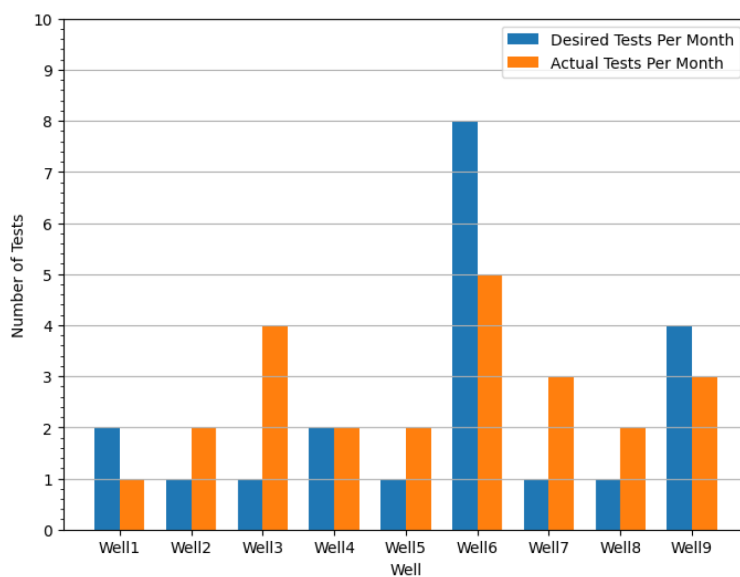


Figure 5—Monthly well tests (actual vs desired) for 9 wells connected to a test separator.

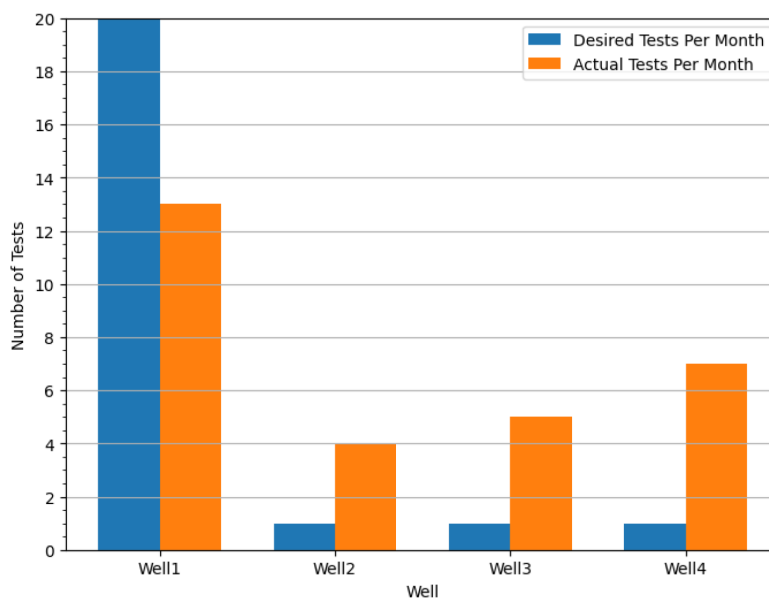


Figure 6—Monthly well tests (actual vs desired) for 4 wells connected to a test separator via an automated header. Well 1 experienced 5 days of downtime.

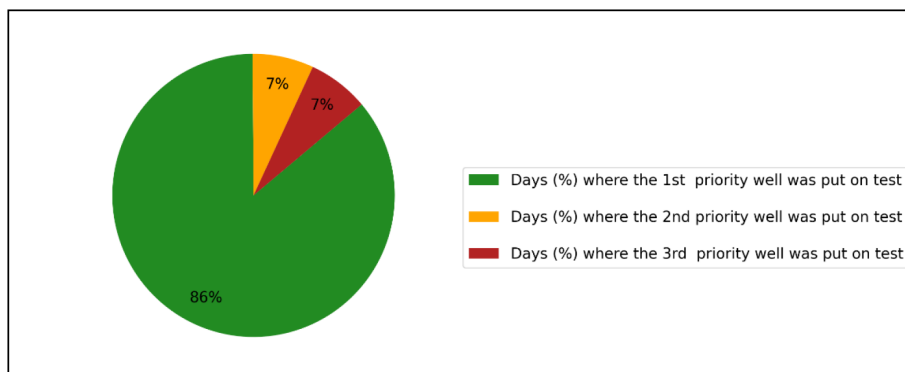


Figure 7—Breakdown of recommendation adoption by well preference

Well Test Validation

Capture of Well-test Records & Auxiliary Measurements

Historically, validation of well tests is performed manually by field operators using a ‘grease’ book where oil, water and gas accumulator values are documented daily and incremental values are used to create a well-test record. This process is prone to human errors in both recording instrument readings and prorating these readings to daily rates. The grease book approach only records basic volumetric information and does not provide for comprehensive analysis of the events that may have taken place over the course of a test (e.g. cycling on the electric submersible pump). To overcome these limitations, a mobile app has been deployed where operators can create, view, or edit well-test records.

To eliminate data entry inaccuracies, field automation and SCADA measurements are integrated with the mobile app: upon the completion of a well test, field automation equipment captures and publishes a well-test record (referred to as a ‘Raw’ well-test record) that provides the well/vessel information, the test duration, and the volumes produced during the test, among other fields. The raw well-test record is the source record for each individual well test and can be visualized through the mobile app. Once a raw well-test record is validated by the field personnel, recorded volumes are automatically prorated to daily rates and the test is approved for reporting to state/federal agencies and utilized for liquid volume allocation.

Processing & Validation of Well-Tests

The objective of the Well-Test Validation workflow (WTV) is to establish a systematic and automated method of validating production well tests. WTV was developed as an expert system based on proprietary rules developed for assessing the quality of well tests of Permian Basin assets. In total, more than 60 rules are used to validate a well-test record utilizing well-test field records, well information (e.g. producing method), and high-frequency SCADA data (e.g. surface, downhole pressure, volume accumulators). The workflow does not seek to modify or edit the recorded oil, water, or gas volume but rather to ensure that the raw well-test record accurately and reliably represents the well's unimpeded flowing conditions at the time of the test, with emphasis on separation efficiency and continuous and uninterrupted production.

The Well-Test Validation workflow applies the predefined rules and thresholds to evaluate each raw well test, categorizing the results as ‘Valid’, ‘Invalid’, or ‘Alert’. Each rule can escalate a test from **Valid** to either an **Alert** or **Invalid** status based on the severity of the violation. Tests marked as **Valid** are accepted as official records and used for production allocation. In contrast, **Invalid** tests are excluded from allocation but retained for informational purposes. Tests flagged as **Alert** are considered ‘Pending’ and require further review by field personnel to determine if they should be trusted as representative well tests.

To facilitate timely action, the results are communicated to the field through a mobile application, supported by a notification system that ensures prompt attention to pending well tests. Figure 8 illustrates the logic diagram underpinning the well-test validation process. Additionally, WTV provides concise diagnostic feedback to assist operators in making informed decisions to either approve or reject a well-test record. It is important to note that field operators retain visibility and edit access to both Valid (auto-approved) and Invalid (auto-rejected) records. Figures 9A and 9B present examples of the mobile interface for tests classified as Valid and Alert. Figure 9C presents the notification for a test with pending approval status.

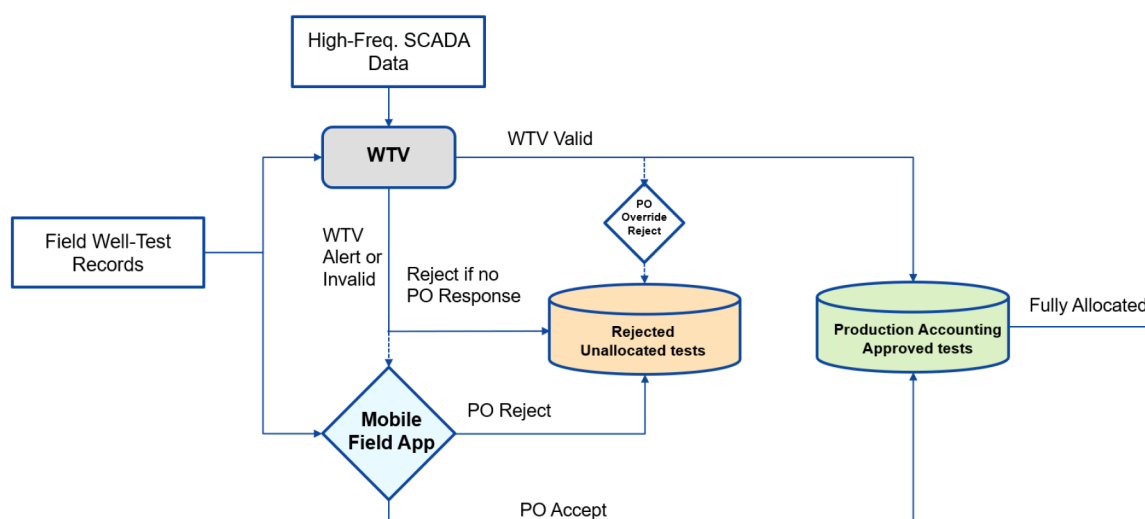


Figure 8—Logic diagram for well-test approval process, inclusive of field Production Operator (PO) actions

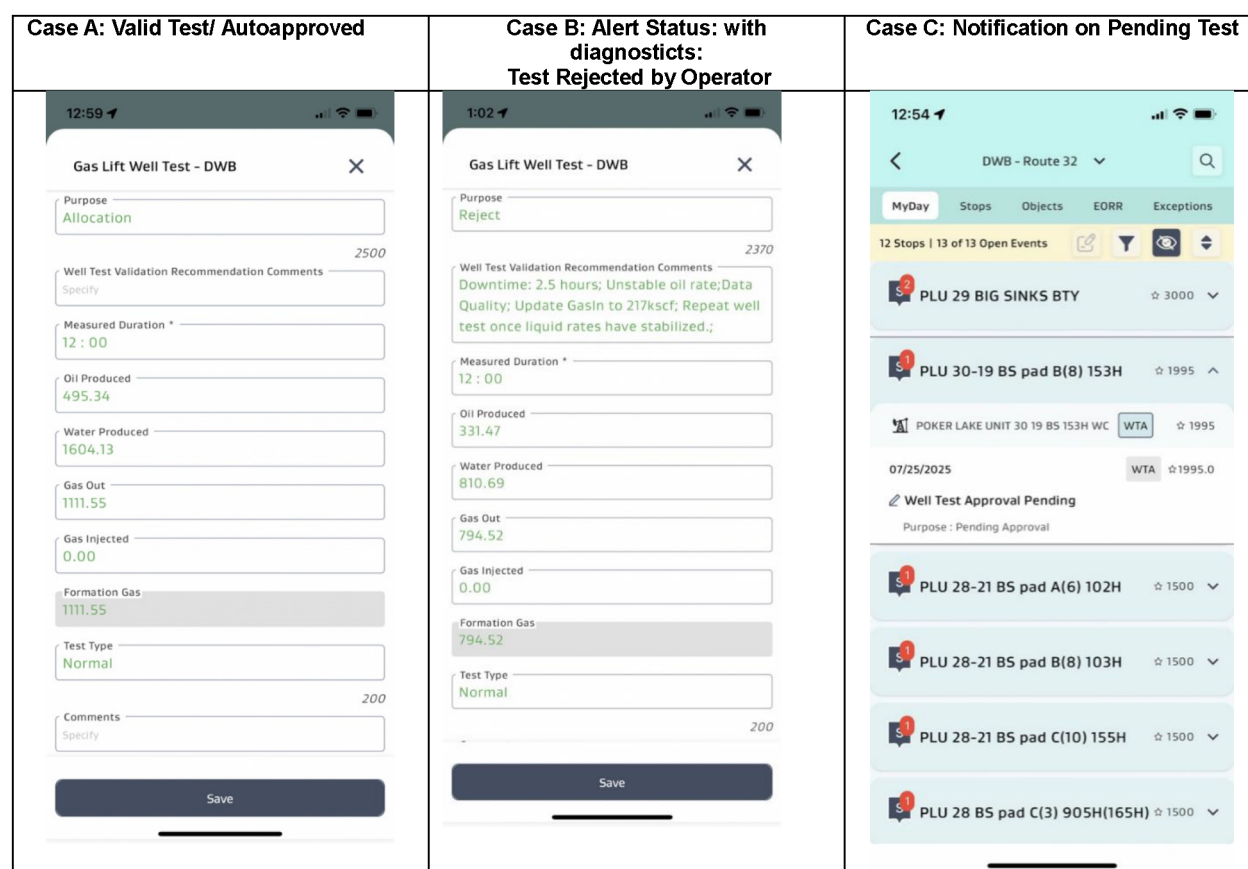


Figure 9—Integration of Well-Test Validation with Mobile App for manual approval of well tests. Illustrated examples for Valid (9A), Alert (9B) and Pending records (9C). Units in STBO and kScf.

The evaluation of a well test is initiated shortly after its completion, typically within an hour, once high-frequency SCADA data becomes available after the test period. This represents a substantial improvement in well-test processing time compared to the manual approval process which often extended several days. Under the current automated configuration, approximately 60% of tests are auto-approved (classified as Valid), 30% are flagged for review (Alert), and 10% are auto-rejected (Invalid). This automation has enhanced consistency in well-test validation and significantly reduced the time field personnel spend on

routine data screening. As a result, operational focus has shifted toward higher-value activities such as artificial lift troubleshooting, flowline routing, separator maintenance, and production debottlenecking. Furthermore, the integrated notification system ensures timely action on all pending well-test records, thereby maximizing the capture and utilization of rate data from the field. This marks a departure from the earlier approach, which prioritized regulatory compliance and often left surplus data underutilized.

The Well-Test Validation workflow incorporates functionality to detect well downtime by leveraging high-frequency data from both the wellhead and separator such as surface and downhole pressures, as well as test separator volume accumulation. A detailed description of the Downtime Advisory workflow is provided by Yan et al. (2024). When downtime is detected, WTV estimates the average flowing pressure (both surface and downhole) and well control parameters (e.g. average choke position or average gas lift injection rate) during the effective flow period of the test. Simultaneously, the produced volume is accurately prorated based on this effective flow duration.

These outputs are integrated to create a comprehensive dataset that correlates flowing pressures and well controls with production rates, forming a robust foundation for machine learning applications in virtual flow metering and production optimization under operational constraints. The effectiveness of such models has been demonstrated in studies by Movahed et al. (2024), and Nallapusala et al. (2025), showing production uplifts of up to 3–4%.

Well-Test Validation Rules

The set of rules used to validate well tests was collaboratively developed by production engineers, measurement specialists, and technology development teams while incorporating aspects from the American Petroleum Institute well-testing guidelines. These rules are regularly reviewed and updated to refine their scope and improve accuracy. Each rule leverages high-frequency data and compares a computed functional output against predefined Alert and Invalid thresholds. If any single rule is outside the Invalid threshold, the entire test is assigned an Invalid status. Tests with all of the raw measurements falling inside of the first threshold are marked as Valid and automatically accepted as official records used for production allocation; Invalid tests, those with a single rule outside the Invalid threshold, are automatically excluded from allocation, but retained for informational purposes, with the final assignment subject to change by the operator.

The implemented rules are divided into five broad categories:

Data Quality Rules – Checks for discrepancies between the well's producing method and actual measurements, missing or out-of-range measurements, flatlines, and negative volume accumulator trends. If any of those issues are detected, relevant diagnostics are passed to the field personnel via the mobile app. Not all data quality rules will invalidate a test – some of them serve for information only. For example, a failed downhole pressure gauge will not invalidate a well test.

Well Operation Rules – Checks for shut-ins during the well test, Electric Submersible Pump (ESP) downtime, ESD (Emergency Safety Device / valve) events, significant choke or gas-lift changes, severe slugging, or unstable liquid rate.

Separation Efficiency Rules – Checks that oil, water, and gas phases are sufficiently separated based on Coriolis oil density and drive gain measurements.

Volume and Rate Validation Rules – Ensures the totalizer volumes captured by field automation are consistent with the volumes inferred from high-frequency separator sensor data (e.g. oil, water and gas accumulation and net-gas volumes).

Production Deviation Rules – Ensures that the totalizer oil, water and gas rates are in line with prior well tests and in agreement with the current well controls. These checks aim to minimize rate variance across well tests.

The section below presents illustrative examples of Well-Test Validation rules.

Example 1: Well downtime during test

To accurately capture well rates during a production well test, oil, water and gas volumetric flow need to be continuous and stabilized for the duration of the test. Figure 10 illustrates a 24-hour production well test, inclusive of a 9-hour downtime event comprising a choke-back period followed by an emergency shut-down event. The downtime event can be detected from pressure and volume accumulation measurements (i.e. no accumulation of volume) and confirmed by the signals from the ESD device. Due to the shut-in event, the test is not considered representative and hence not auto-approved for allocations or regulatory reporting. In such cases (e.g. ESD events or ESP downtime), operators are provided with relevant diagnostics that justify the recommendation.

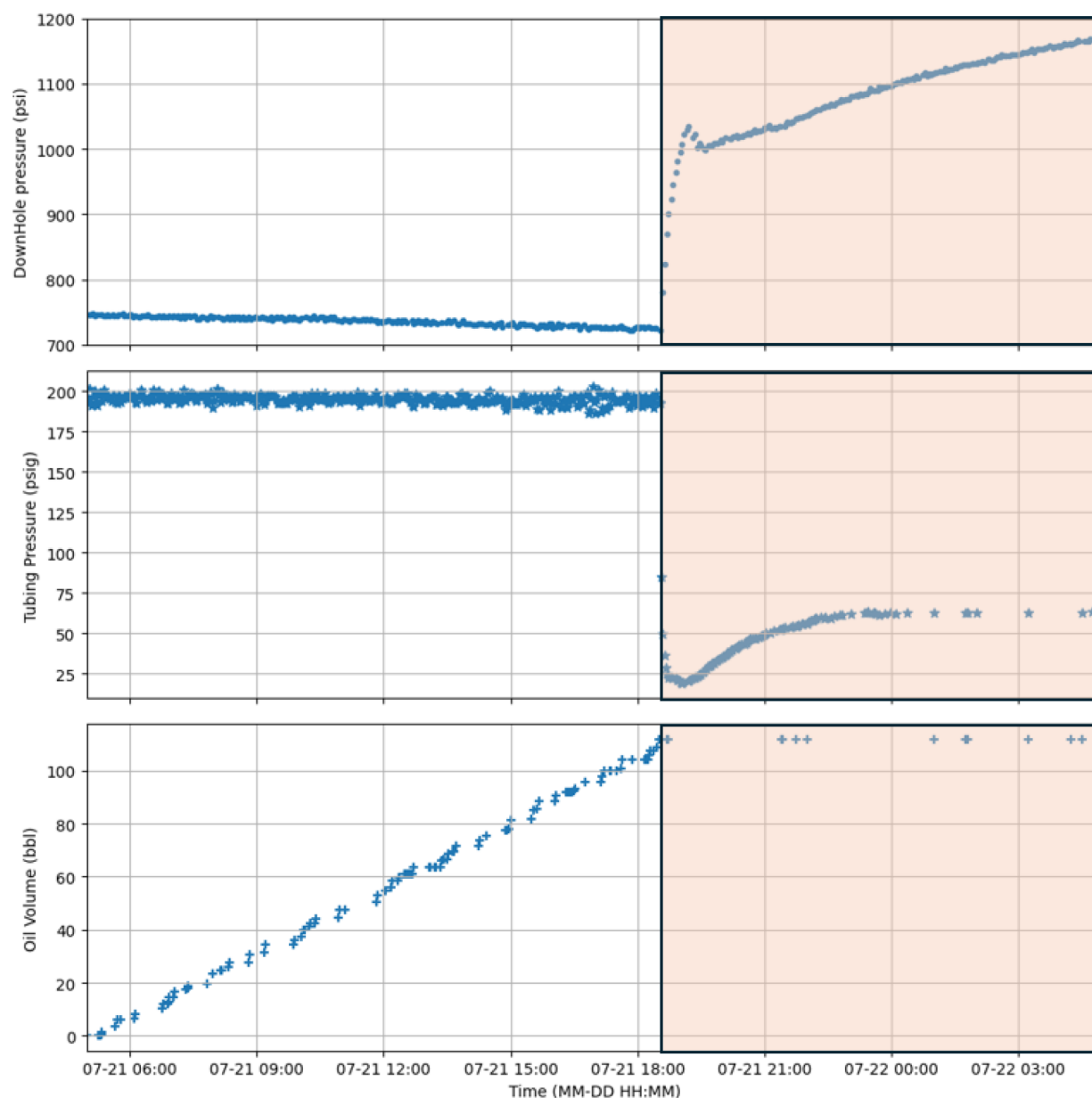


Figure 10—Production well test with shut-in event (9 hours) during the test

Prior to the shut-in event, and for a period of 14 hours, the well test shows stabilized oil production, downhole and tubing pressure measurements. During this interval, the well-test validation system estimates average surface and downhole flowing pressures, well control parameters (e.g., gas lift injection rate or ESP

metrics), and records these alongside oil, water, and gas flow rates. Although the test is inconclusive for allocation purposes, the captured data—linking pressure behavior, control settings, and production rates—is archived for use in machine learning models such as virtual flow metering applications aimed at optimizing artificial lift performance outside of formal well-testing conditions.

This approach enhances the value of production well testing by maximizing data extraction, regardless of the well-test outcome, thereby improving field efficiency and equipment utilization.

Example 2: Temporary re-routing of the well-in-test

Figure 11 depicts a 24-hour well test during which oil and water accumulation remains flat for approximately four hours. During this interval, well pressures are steady but slightly lower, suggesting that the well continued to flow and was likely rerouted to the production separator.

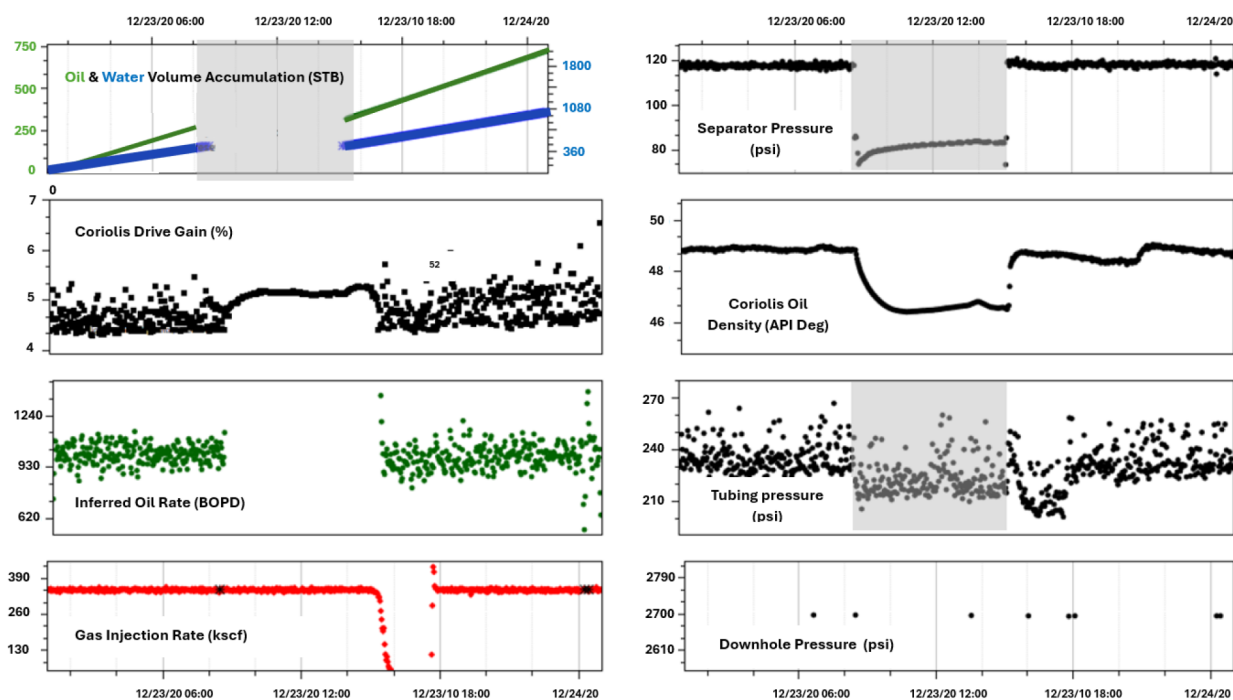


Figure 11—Production well test with temporary re-routing of the well in test.

Since volume accumulation was not observed continuously throughout the full 24-hour duration, the well test was not automatically approved for allocation purposes. Operators were notified of the temporary rerouting and instructed to repeat the well test to ensure data integrity.

This example highlights the importance of continuous surveillance during well testing to ensure accurate volumetric and rate measurements. Similar validation logic is applied to detect premature or delayed well routing at the beginning or end of a well test, reinforcing the need for robust monitoring protocols.

Example 3: Water carry-over during well test

During a steady-state production well test, the separation efficiency between the oil and water phases are monitored using the Coriolis oil density measurement. The Coriolis meter provides an estimate for the density of the mixture flowing through the oil leg, and, for normal separation conditions, is expected to be in line with oil density values, typically 35 API degrees or higher. If the measured oil density is less than 25 API degrees, this is indicative of water carry-over, pointing to poor separation conditions. Figure 12 illustrates a 24-hour well test where the oil density, initially at 47 API degrees, dropped to an average value of 13 API degrees, indicating possible water in the mixture passing through the oil leg and hence poor separation. Brine has an equivalent API density of approximately 3.5 degrees indicating that measured API

gravity values less than 10 may be encountered during water carry over periods. In this example, the water rate measurement is low, the oil rate measurement is high, and the oil leg density is low due to the likely presence of water. For this example, both the duration and severity of the carry-over rate was significant, and the well test was categorized as ‘Invalid’. The Well-Test Validation workflow will detect water carry-over events and, depending on state regulations (e.g. New Mexico or Texas) will ‘Alert’ or ‘Invalidate’ the well test, providing appropriate diagnostics to field operators to inspect the separator and adjust liquid levels.

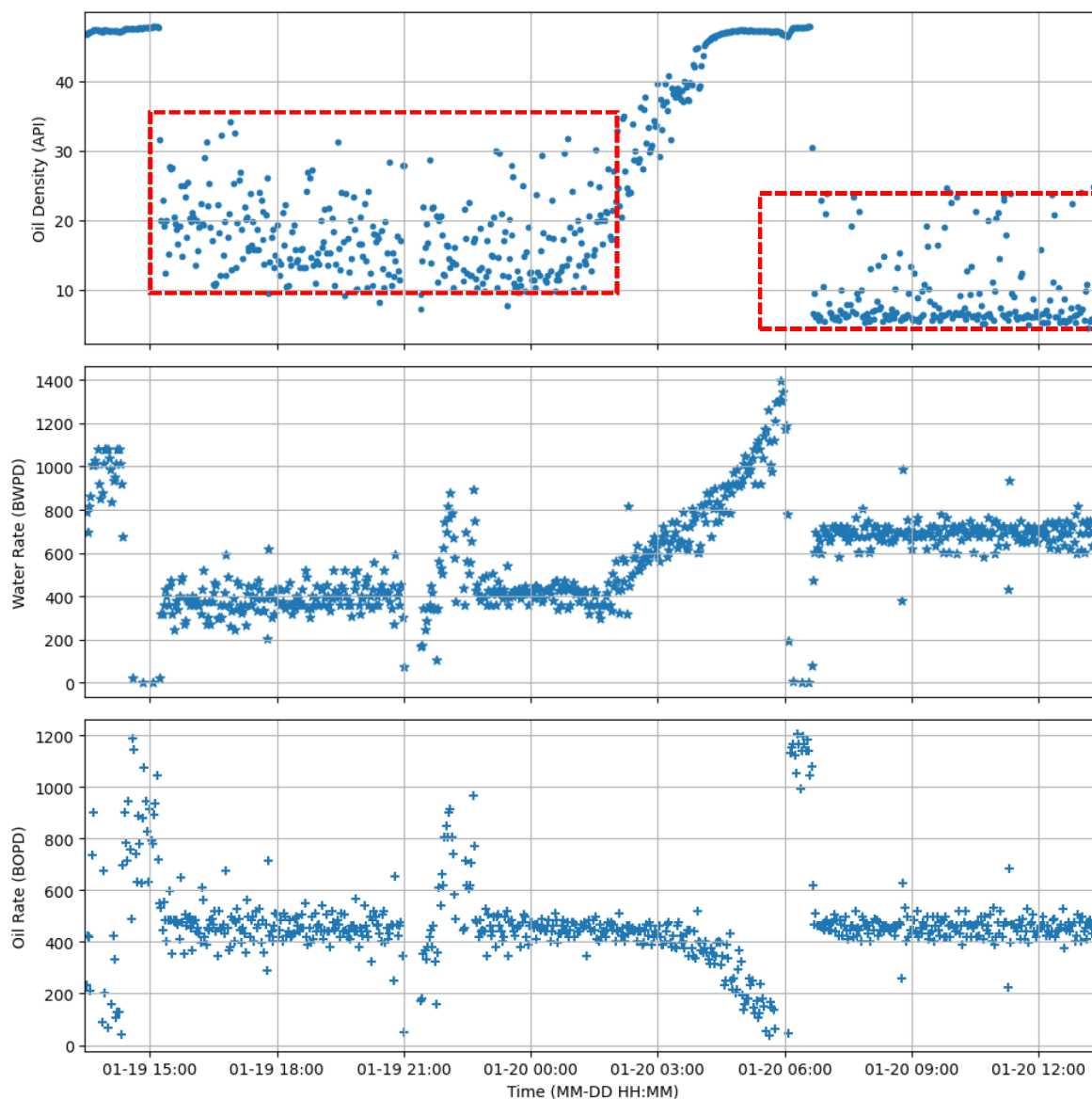


Figure 12—Production well test with high oil density, indicative of water carry-over.

Consecutive ‘Alert’ and ‘Invalid’ recommendations (see Table 1), despite operations efforts to adjust water-oil levels, is indicative of an internal separator issue. Inspection of the test separator, in one instance, revealed frac-sand and other deposits had obstructed the water outlet resulting in the water phase exiting through the oil-leg (Figure 13). This finding demonstrates the capability of the Well-Test Validation workflow to proactively identify equipment issues and support field operations in maintaining well-test integrity.



Figure 13—Identified debris obstructing the separator water outlet

Table 1—Example of persistent alerts/invalid records due to high oil density for a particular test separator.

TestDate	WTVRecommendation	WTVDiagnostics	WTVRecommendations
03/14/21	Valid		
03/13/21	Valid		
03/12/21	Valid		
03/11/21	Invalid	Questionable (negative) net gas volume	Well Test Scheduling / Routing; Unstable
03/10/21	Invalid	High oil density reading from Coriolis	High Oil Density; Production Deviation;
03/09/21	Alert	Questionable (negative) net gas volume	Verify gas injection/production volumes.
03/08/21	Alert	High oil density reading from Coriolis	High Oil Density; Production Deviation;
03/07/21	Alert	High oil density reading from Coriolis	High Oil Density; Production Deviation;
03/06/21	Alert	High oil density reading from Coriolis	High Oil Density; Check separator operat
03/05/21	Invalid	High drive gain reading from Coriolis	High Drive Gain; High Oil Density; Check
03/04/21	Alert	High oil density reading from Coriolis	High Oil Density; Production Deviation;
03/03/21	Invalid	High oil density reading from Coriolis	High Oil Density; Check separator operat
03/02/21	Invalid	High drive gain reading from Coriolis	High Drive Gain; High Oil Density; Well T

Example 4: Production Deviation and well-to-vessel routing verification

Prior to auto-approving an otherwise good-quality well test, oil, water and gas measurements are compared to the most recent approved well test, declined to current conditions. If the deviation is greater than 30% and the change in liquid rate is not justified by variations in well controls then the well test is not auto approved for allocation. An interesting case arises when the measured liquid rate is incompatible with the downhole pressure recorded during the test.

Table 2 illustrates the test history of an unconventional well. The well test on 06/12/2025 recorded an abnormally higher liquid rate yet the flowing pressure was in line with prior well tests. This discrepancy is quantified using a productivity-index approach where the change in liquid rate is divided by the change in the average flowing downhole pressure from recently accepted well tests. If the calculated value of the productivity index is outside of expected bounds (e.g. $|PI| > 20$ BLPD/psi), this is indicative of a non-representative well test, possibly due to an erroneous mapping (e.g. Well A was physically routed to the test separator but Well B was assigned on test). In the example shown in Table 2, a recommendation was

sent to the field operators on 06/12/2025 to validate the well-on-test, which confirmed the mapping error. It is important to note that this approach does not apply to wells incurring frac-hits where both the liquid production and downhole pressure may increase. This example underscores the critical role of downhole pressure measurements in maintaining data integrity during production well testing.

Table 2—Production well-test history of an unconventional well

Date	Test Liquid Rate (BLPD)	Average flowing DHP (psi)	WTV Recommendation	WTV Diagnostics
06/12/2025	815.19	1,060.37	Alert	Possibly wrong well in test; Confirm well X was in test. If not, reject test.
06/11/2025	318.87	1,047.79	Valid	
06/08/2025	338.73	1,145.69	Valid	
06/02/2025	328.26	1,056.04	Valid	

Results and Observation

The implementation of WTV has markedly improved the acceptance rate of well tests, increasing asset well-tests acceptance rate from 65% to 75%. Systematic approval of representative tests and automated rejection of unrepresentative tests ensures consistency. Integrating WTV with a mobile application and allowing operators to focus on the 30% of tests that require manual evaluation has accelerated the validation duration from days to hours. This increased efficiency enables timely input for production optimization, notifies field personnel of critical equipment maintenance (separator clean-outs), and supports real-time surveillance across engineering teams.

WTV has also enabled the development of a comprehensive dataset linking flowing pressures and well-control parameters (e.g., gas lift rates, ESP frequencies) to validated production rates. This dataset forms the foundation for advanced applications such as virtual flow metering, exception-based surveillance (e.g., productivity index monitoring, downtime detection), and machine learning tools (Movahed et al. (2024), Spivey et al. (2025)). Importantly, the data acquisition process ensures that information from all well tests—regardless of approval status—is captured, enabling future workflows to more easily create value. The combined application of the scheduling and validation workflows positively impacts data collection and utilization of field equipment.

Figure 14 illustrates the impact of WTV on a Permian Basin well. This well was routed to a dedicated test separator vessel (this test separator is only used for this specific well), for which field automation was capturing and publishing well-test records on a daily basis. Prior to the enablement of WTV, a portion of well tests were approved for allocation prioritizing regulatory reporting (black dots). Following WTV implementation, all valid tests were approved. We observe that the oil allocation (green line) is smoother in the post implementation period suggesting improved and more accurate liquid volume allocations.

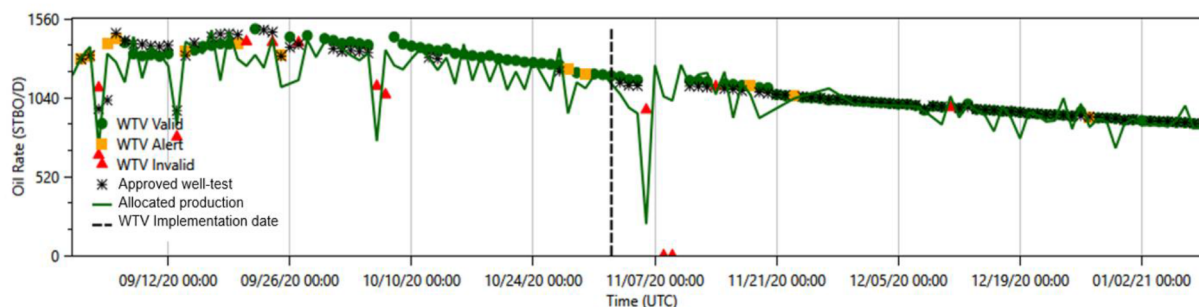


Figure 14—Example of test validation before and after the implementation of the well-test validation.

Since the initial deployment of the Well-Test Validation (WTV) workflow in 2020, regular field training and proactive identification of separator-related issues have been instrumental in securing user buy-in and driving rapid adoption. As a result, the workflow has been successfully rolled out across the Permian Basin, covering over 2,300 wells at the time of this publication. Field operators retain full visibility and control over all well-test records, with the ability to override WTV recommendations, when necessary. Historically, only 4% of auto-processed records are overwritten—such as cases where WTV-valid tests are manually rejected—providing valuable feedback to refine the expert system. As illustrated in Figure 15, approximately 50% of ‘Pending’ tests (i.e., those flagged with an Alert status) are manually approved by operators, indicating that these cases lack sufficient evidence for automated validation and require human judgment.



Figure 15—Comparison of WTV recommendations with final test status (approved/rejected).

Discussion & Future Work

The need for high-quality well-test data was highlighted during development of advanced surveillance and optimization workflows such as automated gas lift optimization (Movahed et al. (2024)) and downtime identification (Wang et al. (2024)). By developing WTV and effectively training field staff in its use, we were able to obtain high uptake and rapid deployment. Rapid uptake was accelerated by providing operators with timely and high-quality well-test recommendations on their mobile app. Uptake was also enhanced by early identification of potential separator or measurement problems with proposed actions, as discussed previously with the identified debris blocking the outlet (Figure 13). While WTV has improved well-test acceptance rates, ongoing stewardship is maintained through regular meetings and surveillance dashboards to ensure that all ‘Pending’ tests (i.e., those flagged with ‘Alerts’) are promptly reviewed and resolved by field personnel.

Further opportunities to enhance well-test quality have been identified including optimizing the duration of well tests and the number of wells per test separator, as well as managing test and bulk separator pressure differences. Currently, the ability to optimize well-test durations is constrained by state and federal regulations. For instance, increased flexibility in well-test durations could enable the concept of well segmentation, whereby a valid flow period could be extracted from an otherwise invalid test. This approach would allow the definition of a "virtual valid test" from a stable subset of a well-test resulting in improved data utilization. In facilities with limited automation, well-to-separator mappings and test timing (start/end) can be captured via the mobile application instead of relying on Human Machine Interface (HMI) devices. This shift could reduce dependency on costly automation hardware and simplify installation procedures. Finally, deploying the well-test scheduling workflow on automated headers—combined with dynamic test durations—could significantly increase well-to-tester ratios. This would result in substantial savings in facility and separator equipment requirements, while enhancing operational efficiency. Separator pressure

differences between test and production separators remains a challenge, and further efforts are required to modernize facilities using Pressure Control Valves on test and production vessels.

Conclusions

This paper presents a novel, integrated workflow for well-test scheduling and validation that significantly enhances operational efficiency and data quality. The technology streamlines well-test scheduling and provides time savings via the automated and systematic approval and/or rejection of well tests, ultimately improving data collection both quantitatively (more tests) and qualitatively (higher-quality results). The workflow also provides daily insights to field personnel on the performance and health of test separators and other measurement equipment, enabling timely detection and resolution of issues. Furthermore, the validated well-test data serves as a reliable training dataset for future AI/ML-based optimization and surveillance workflows, empowering field engineers with more accurate production data to support critical business decisions in production optimization and field development planning.

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